Study Planner — Demo

Team T-W3 · CS252 Group 02 · captured from the real CLI on the `main` branch

What it does

The planner reads tasks from a CSV file, ranks them by importance and deadline, fills the available study hours greedily, and prints the plan. The CLI in `shared/study_planner/__main__.py` chains four stages:

```
parse_csv  →  prioritise  →  allocate  →  render
```

Input

`samples/tasks.csv`:

```
title,importance,effort_h,deadline
DS exam,5,6,2026-10-30
Assignment,4,1.5,2026-10-27
```

All runs below pass `--today 2026-10-26` so the output is reproducible; without it the planner uses the current date.

1. Basic run — 2 hours

The Assignment is due first, so it gets its full 1.5 h; the remaining 0.5 h goes to the DS exam.

```
$ python -m shared.study_planner samples/tasks.csv --hours 2 --today 2026-10-26
Study plan:
- Assignment: 1.50 h
- DS exam: 0.50 h
Total: 2.00 h
[exit code 0]
```

2. Enough time for everything — 10 hours

Every task fits, so each gets its full effort and 2.5 h stay unused.

```
$ python -m shared.study_planner samples/tasks.csv --hours 10 --today 2026-10-26
Study plan:
- Assignment: 1.50 h
- DS exam: 6.00 h
Total: 7.50 h
[exit code 0]
```

Error handling

Bad input stops the planner with an error and a non-zero exit code. Tracebacks are shortened to their final line here.

3. Missing file

```
$ python -m shared.study_planner missing.csv --hours 2
Traceback (most recent call last):
  ...
FileNotFoundError: Task file not found: missing.csv
[exit code 1]
```

4. Empty file

```
$ python -m shared.study_planner empty.csv --hours 2
Traceback (most recent call last):
  ...
ValueError: The task file is empty: no tasks found.
[exit code 1]
```

5. Infeasible plan (InfeasiblePlan)

```
$ python -m shared.study_planner samples/tasks.csv --hours 0 --today 2026-10-26
Traceback (most recent call last):
  ...
shared.study_planner.errors.InfeasiblePlan: Plan infeasible: needed 0 h, but only 0 h
    available.
[exit code 1]
```

6. Malformed row

`samples/tasks_bad.csv` has `importance = high`.

```
$ python -m shared.study_planner samples/tasks_bad.csv --hours 2
Traceback (most recent call last):
  ...
ValueError: invalid literal for int() with base 10: 'high'
[exit code 1]
```

Running the tests

```
python -m pytest tests/test_study_planner_e2e.py
```

See `shared/study_planner/README.md` for setup and the CSV format.